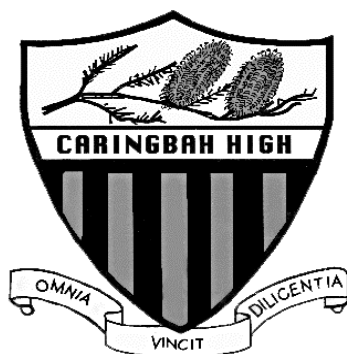




Caringbah High School

Year 11 2017

Mathematics Extension 1

Preliminary Course

Semester 2 Exam

General Instructions

- Reading time – 3 minutes.
- Working time – 90 minutes.
- Write using black or blue pen.
- Board-approved calculators may be used.
- Relevant mathematical reasoning and/or calculations **MUST** be shown.
- Marks may not be awarded for answers which do not show relevant working and/or reasoning.
- A reference sheet is provided.

Total marks – 65

Write your answers in the answer booklets provided.

Start each question in a new booklet.

Ensure your name is clear.

Name: _____

Class: _____

Question 1: (13 marks)**Marks**

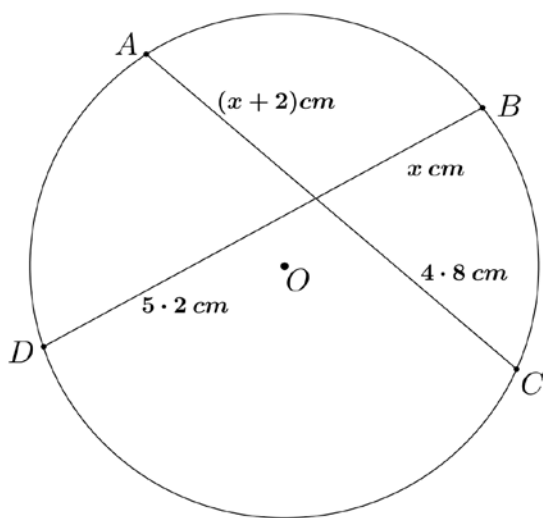
- a) Solve $x - 8 \leq \frac{20}{x}$. 3
- b) Sketch the graph of the function $y = 3^{x+1}$ 2
- c) Find the exact value of $\cos 165^\circ$. 1
- d) Find the derivative of the following expressions, expressing your answer in simplest form.
- i) $5x^3 - 3x^2 + 2$ 1
- ii) $\sqrt[3]{x}$ 1
- iii) $(5x - 2)^3$ 1
- iv) $(3x + 1)(2x - 3)^4$ 2
- v) $\frac{2x + 1}{\sqrt{x - 3}}$ 2

Question 2: (13 marks)**Marks**

a) Factorise $p^6 + 8$ 1

b) Let $f(x) = 2x^2 - 3x$. Find $f'(x)$ by differentiating from first principles. 2

c) Find the value for x , giving reasons. 2



d) The point $B(5, -2)$ divides the interval joining the points $A(3, 7)$ and $C(x, y)$ internally in the ratio $2:3$.

Find the coordinates of point C .

e) Sketch $y = \frac{x+1}{x-2}$, clearly showing any asymptotes and intercepts on axes. 3

f) Solve $\sec^2 \theta = 7 \tan \theta - 1$ for $0^\circ \leq \theta \leq 180^\circ$, to nearest degree. 3

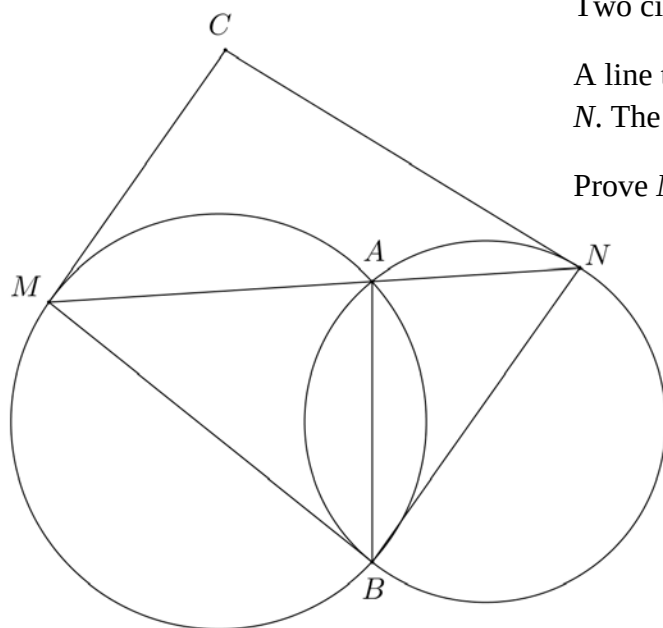
Question 3: (13 marks)**Marks**

a) State the domain and range of the graph of $y = \sqrt{x-9}$. 2

b) Find the equation of the straight line that passes through the intersection of the lines $x + 3y - 2 = 0$ and $2x - y - 5 = 0$ and also passes through the point $(5, -2)$. Express your answer in general form. 3

c) Find the values for k such that the line $12x + 5y + k = 0$ is a tangent to the circle $(x-1)^2 + (y+5)^2 = 9$. 3

d) 3



Two circles intersect at A and B .

A line through A cuts the circles at M and N . The tangents at M and N meet at C .

Prove M, N, C and B are concyclic.

e) i) Show that $x + 3 - \frac{4}{x+1} \equiv \frac{x^2 + 4x - 1}{x+1}$ 1

ii) Hence state the oblique asymptote of the graph $y = \frac{x^2 + 4x - 1}{x+1}$ 1

Question 4: (13 marks)**Marks**

a) α and β are roots of the equation $2x^2 - 3x + 5 = 0$. Find the values of:

i) $\alpha\beta$

1

ii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$

2

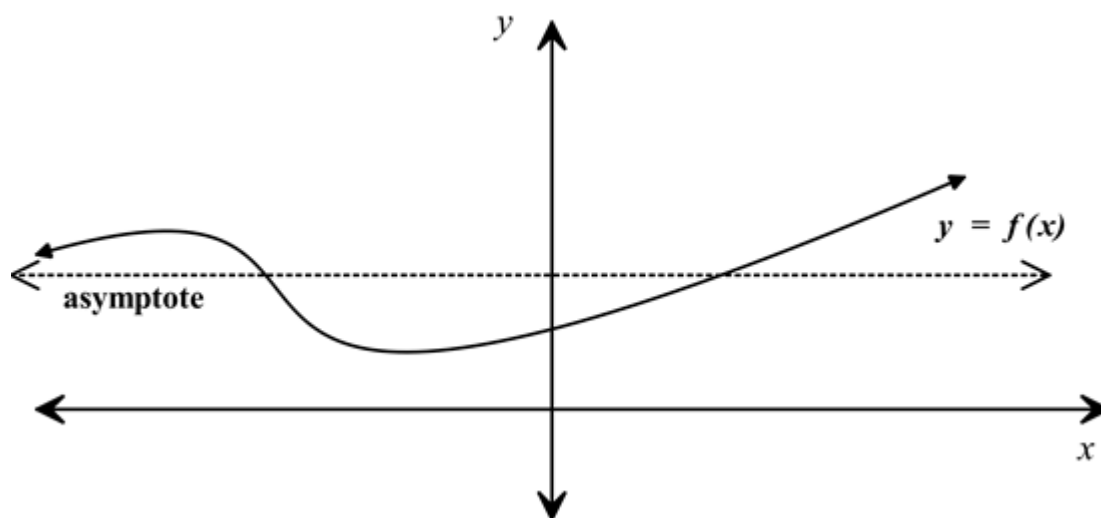
iii) $(\alpha - \beta)^2$

2

iv) $\alpha^3 + \beta^3$

2

b) Copy or trace the function $y = f(x)$ into your answer booklet. Draw a neat sketch of the derivative function $y = f'(x)$, showing its relationship to $y = f(x)$.

3

c) Shade the region on the Cartesian number plane where the following inequalities hold simultaneously.

3

$$x^2 + y^2 < 4 \text{ and } -1 < x \leq 2.$$

Question 5: (13 marks)

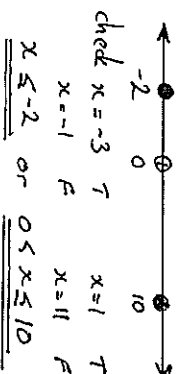
- a) Find the point that divides the interval between the points $A(-3, -1)$ and $B(5, 9)$ externally in the ratio 5:3. 2
- b) Find the equation of the normal to the curve $y = x^3 - 2x^2 + 4x + 1$ at the point $(2, 9)$. 3
- c) i) By solving simultaneously, show that the two parabolas 2
 $y = 4x^2 + 6x - 1$ and $y = x^2 + 2x + 3$ intersect at the point $(-2, 3)$.
- ii) Find the acute angle between these two parabolas at this point of intersection. 2
Express your answer correct to the nearest minute.
- d) Express $x^4 - 2x^3 + 9x^2 - 8x - 20$ in the form $(x^2 - x)^2 + b(x^2 - x) + c$. 4
Hence solve $x^4 - 2x^3 + 9x^2 - 8x - 20 = 0$.

End of Exam

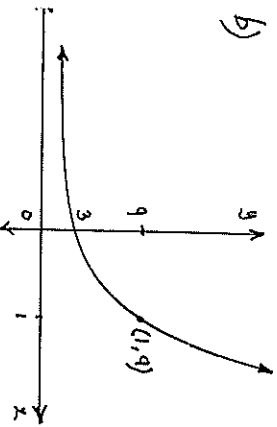
2017 Maths EMT 1
YEAR 11 Sem 2

Question 1

a) $x \neq 0$ $x^2 - 8x - 20 \leq 0$
 $(x-10)(x+2) \leq 0$



check $x = -3$ T $x = 1$ T
 $x = -1$ F $x = 11$ F
 $x \leq -2$ or $0 < x \leq 10$



c) $\cos 165^\circ = \cos 120^\circ \cos 45^\circ - \sin 120^\circ \sin 45^\circ$
 $= -\frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$
 $= -\frac{1+\sqrt{3}}{2\sqrt{2}}$

d) i) $f'(x) = 15x^2 - 6x$
ii) $f'(x) = \frac{1}{3}x^3$ or $\frac{1}{3\sqrt{x^2}}$

iii) $f'(x) = 15(5x-2)^2$
iv) $f'(x) = 8(3x+1)(2x-3)^3 + 3(2x-3)^4$
 $= (2x-3)^3(24x+8+6x-9)$
 $= (2x-3)^3(30x-1)$

v) $f'(x) = (x-3)^{\frac{1}{2}} \times 2 - (2x+1) \times \frac{1}{2}(x-3)^{-\frac{1}{2}}$
 $= \frac{1}{2}(x-3)^{\frac{1}{2}} [4(2x-3) - (2x+1)]$
 $= \frac{x-3}{2(2x-3)^{\frac{3}{2}}}$

Question 2

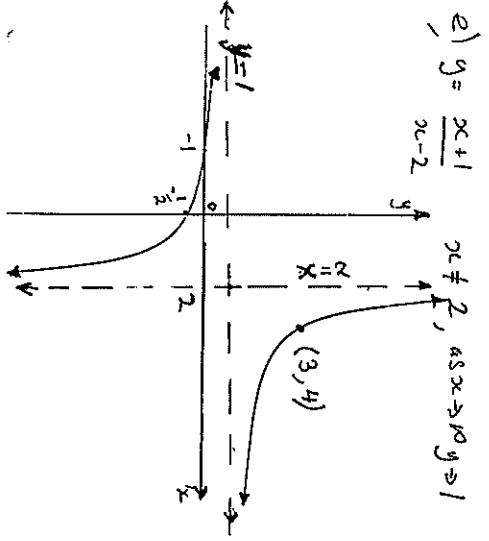
a) $p^6 + 8 = (p^2+2)(p^4-2p^2+4)$

b) $f(x) = 2x^2 - 3x$
 $f(x+h) = 2(x+h)^2 - 3(x+h)$
 $= 2x^2 + 4xh + 2h^2 - 3x - 3h$

$f'(x) = \lim_{h \rightarrow 0} \frac{(2x^2 + 4xh + 2h^2 - 3x - 3h) - (2x^2 - 3x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 3h}{h}$
 $= \lim_{h \rightarrow 0} 4x + 2h - 3$
 $= 4x - 3$

c) $(x+2) \times 4 \cdot 8 = 5 \cdot 2x$
(product of intercepts of chords)
 $0 \cdot 4x = 9 \cdot 6$
 $x = 24$

d) $\frac{2x+3}{5} = 5$ $\frac{2y+3}{5} = -2$
 $2x+3 = 25$ $2y+3 = -10$
 $2x = 22$ $2y = -13$
 $x = 11$ $y = -6.5$



f) $\sec^2 \theta = 7 \tan \theta - 1$
 $1 + \tan^2 \theta = 7 \tan \theta - 1$
 $\tan^2 \theta - 7 \tan \theta + 2 = 0$

let $x = \tan \theta$
 $x^2 - 7x + 2 = 0$
 $x = \frac{7 \pm \sqrt{49-8}}{2}$
 $= \frac{7 \pm \sqrt{41}}{2}$

$\theta = \tan^{-1} \left(\frac{7 \pm \sqrt{41}}{2} \right)$

$\theta = 82^\circ$ or 17°

Question 3

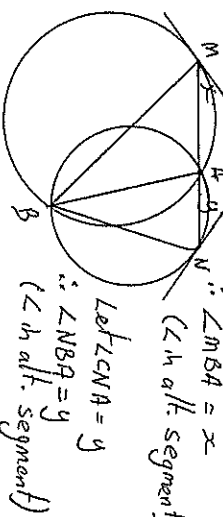
a) D: $x \geq 9$
R: $y \geq 0$

b) $L_1 + k L_2 = 0$ $(5, -2)$
 $(x_1 + 3y_1 - 2) + k(2x_1 - y_1 - 5) = 0$
 $(5 - 6 - 2) + k(10 + 2 - 5) = 0$
 $-3 + 7k = 0$
 $k = \frac{3}{7}$

$(x+3y-2) + \frac{3}{7}(2x-y-5) = 0$
 $7x+21y-14+6x-3y-15=0$
 $13x+18y-29=0$

c) centre $(1, -5)$ radius $= 3$
 $d = \frac{\sqrt{12^2 + 5^2} \cdot |k|}{\sqrt{12^2 + 5^2}}$
 $3 = \frac{|12k-13|}{13}$
 $39 = 12k-13$ or $39 = -12k+13$
 $k = 5$ or $k = -3$

d) Let $\angle CMN = x$
 $\therefore \angle MBA = x$
($\angle$ in alt. segment)



So $\angle MBN = x + y$ (adjacent angles)
 $\angle MCN = 180 - (x+y)$ ($\angle$ sum of $\triangle CMN$)

$\therefore \angle MBN + \angle MCN = 180$
 $\therefore MNCB$ are concyclic.
(opp $\angle$ s of cyclic quad are supplementary)

e) $x+3 - \frac{4}{x+1} = \frac{(x+3)(x+1) - 4}{x+1}$
 $= \frac{x^2 + 4x + 3 - 4}{x+1}$
 $= \frac{x^2 + 4x - 1}{x+1}$

ii) As $x \rightarrow \infty$ $\frac{4}{x+1} \rightarrow 0$
 $\therefore$ asymptote is $y = x+3$

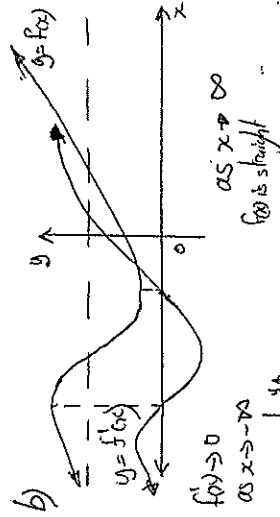
Question 4

a) $\alpha + \beta = \frac{3}{2}$ $\alpha \beta = \frac{5}{2}$

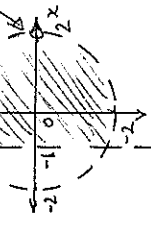
ii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$
 $= \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}$
 $= \frac{(\frac{3}{2})^2 - 2 \times \frac{5}{2}}{\frac{5}{2}}$
 $= -\frac{11}{25}$

$$\begin{aligned} \text{iii)} (\alpha - \beta)^2 &= \alpha^2 + \beta^2 - 2\alpha\beta \\ &= (\alpha + \beta)^2 - 4\alpha\beta \\ &= \left(\frac{3}{2}\right)^2 - 4 \times \frac{5}{2} \\ &= -\frac{31}{4} \end{aligned}$$

$$\begin{aligned} \text{iv)} \alpha^3 + \beta^3 &= (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2) \\ &= (\alpha + \beta)((\alpha^2 + \beta^2) - \alpha\beta) \\ &= \frac{3}{2} \times \left[\left(\frac{3}{2}\right)^2 - 2 \times \frac{5}{2} - \left(-\frac{31}{4}\right)\right] \\ &= -\frac{63}{8} \end{aligned}$$



c) $(2, 0)$ is NOT included.



Question 5

$$1) x = \frac{5 \times 5 - 3 \times 3}{5 - 3} = \frac{5 \times 9 - 3 \times 1}{5 - 3}$$

$$= \frac{25 + 9}{2} = \frac{48}{2}$$

$$P(17, 24)$$

$$\begin{aligned} \text{b)} \quad y &= x^3 - 2x^2 + 4x + 1 \\ y' &= 3x^2 - 4x + 4 \\ x &= 2 \end{aligned}$$

$$m_T = 12 - 8 + 4$$

$$m_T = 8$$

$$m_T = \frac{1}{8}$$

$$\text{Eqn of normal}$$

$$(y - q) = -\frac{1}{8}(x - 2)$$

$$8y - 72 = -x + 2$$

$$x + 8y - 74 = 0$$

$$\text{c) i) } 4x^2 + 6x - 1 = x^2 + 2x + 3$$

$$3x^2 + 4x - 4 = 0$$

$$\frac{(3x + 6)(3x - 2)}{3} = 0$$

$$x = -2 \text{ or } x = \frac{2}{3}$$

$$\text{at } x = -2 \quad y = (-2)^2 + 2(-2) + 3$$

$$= 4 - 4 + 3$$

$$= 3$$

$$\text{ii) } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad y_1' = 8x + 6$$

$$m_1 = 40$$

$$y_2' = 2x + 2$$

$$m_2 = -2$$

$$\tan \theta = \left| \frac{-10 + 2}{1 + 20} \right|$$

$$\therefore \theta = 20^\circ 51'$$

$$\text{d) } x^4 - 2x^3 + 9x^2 - 8x - 20 = (x^2 - x)^2 + b(12x) + c$$

$$\text{Let } x = 0 \quad LHS = -20 \quad RHS = c \quad \therefore c = -20$$

$$\text{Let } x = 2 \quad LHS = 0 \quad RHS = 4 + 2b + c = 0$$

$$\therefore b = 8$$

$$(x^2 - x)^2 + 8(x^2 - x) - 20 = 0$$

$$u^2 + 8u - 20 = 0$$

$$(u + 10)(u - 2) = 0 \quad u = 2 \quad u = -10$$

$$x^2 - x = -10 \quad \text{or } x^2 - x = 2$$

$$x^2 - x + 10 = 0 \quad x^2 - x - 2 = 0$$

$$\text{No solution} \quad x = -1, x = 2$$